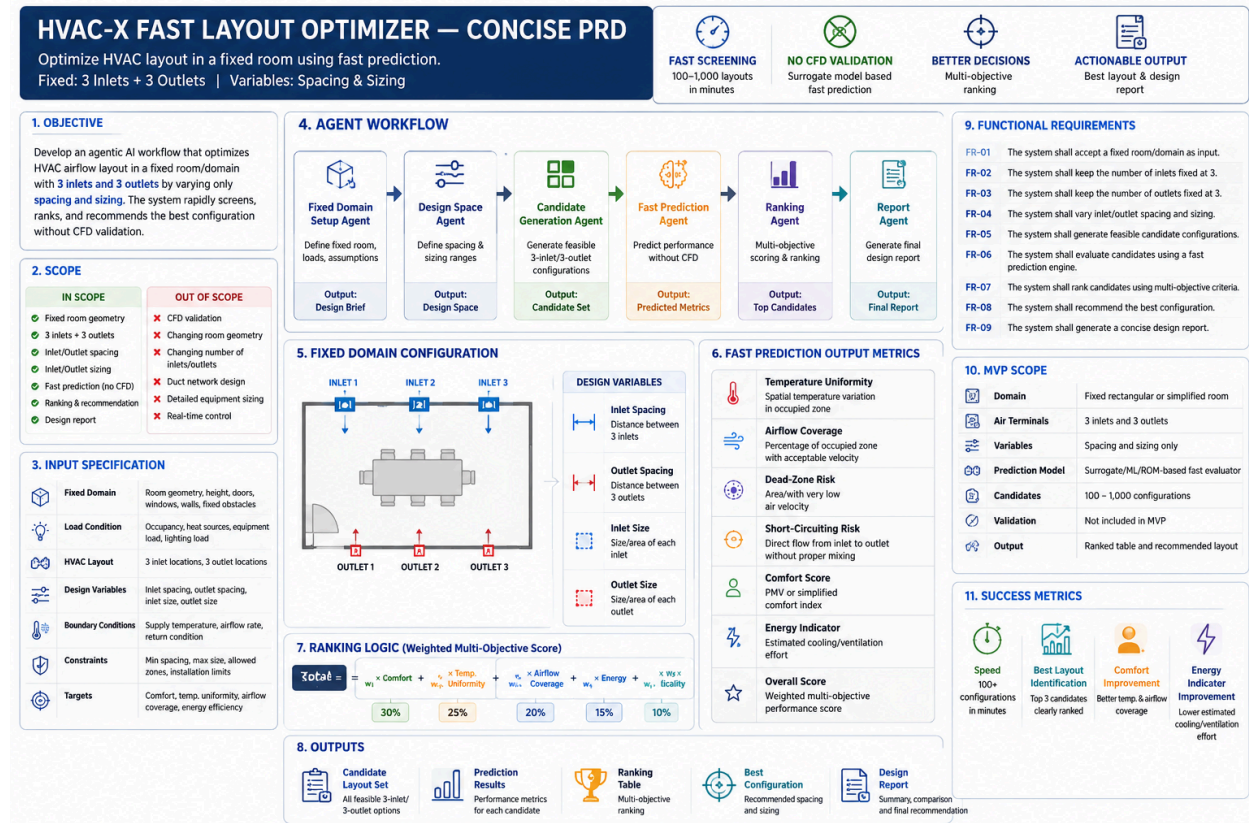




Agentic HVAC Fast Layout Optimizer

1. Product Name

HVAC-X Fast Layout Optimizer

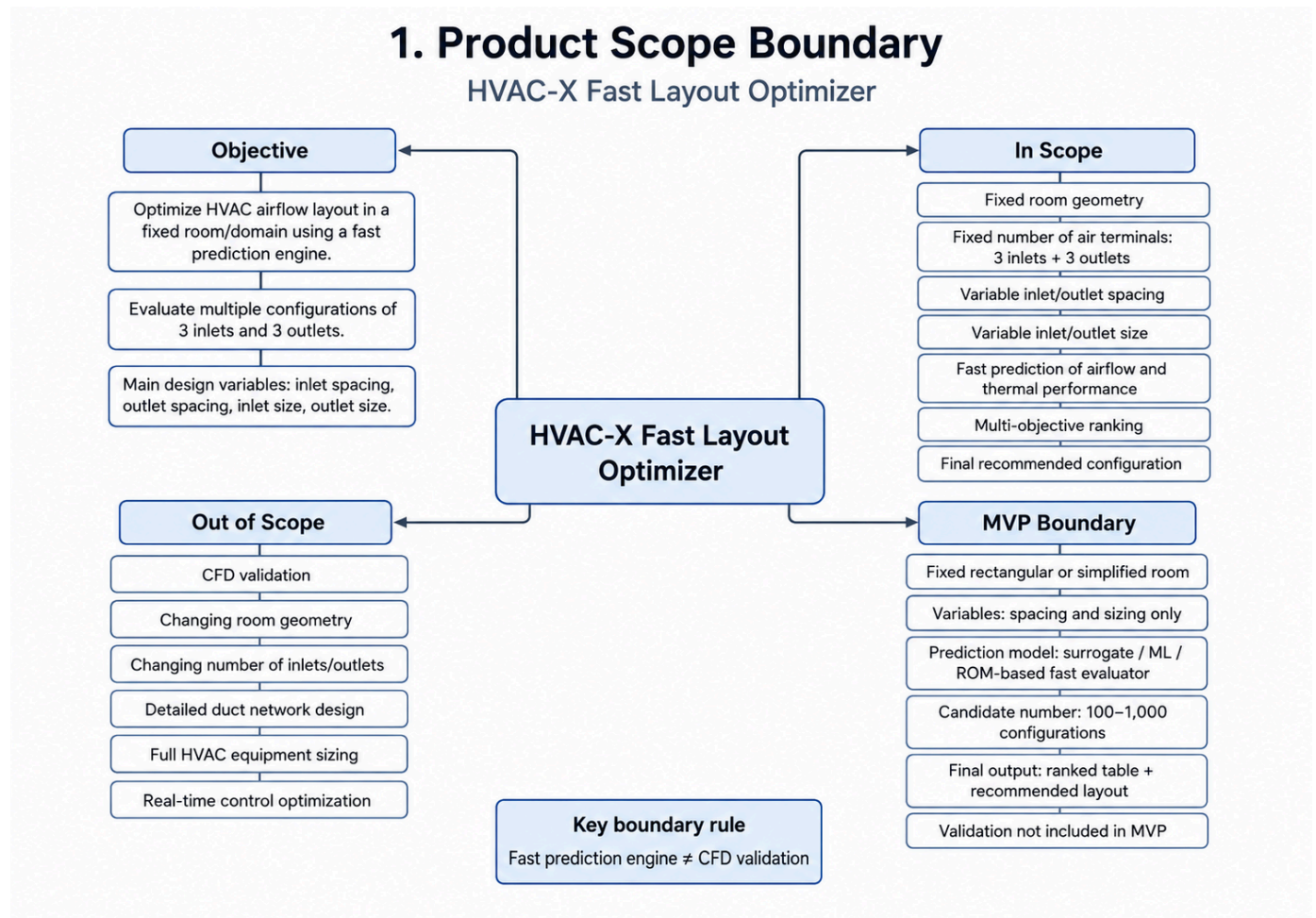
2. Product Objective

Develop an agentic AI workflow that optimizes HVAC airflow layout in a **fixed room/domain** using a fast prediction engine. The system evaluates multiple configurations of **3 inlets and 3 outlets**, where the main design variables are:

- Inlet spacing
- Outlet spacing
- Inlet size
- Outlet size

The system does not perform CFD validation in the MVP. It focuses on rapid screening, ranking, and recommendation based on comfort, airflow quality, and energy-related indicators.

3. Scope



In Scope

- Fixed room geometry
- Fixed number of air terminals: **3 inlets + 3 outlets**
- Variable inlet/outlet spacing
- Variable inlet/outlet size
- Fast prediction of airflow and thermal performance
- Multi-objective ranking
- Final recommended configuration

Out of Scope

- CFD validation
 - Changing room geometry
 - Changing number of inlets/outlets
 - Detailed duct network design
 - Full HVAC equipment sizing
 - Real-time control optimization
-

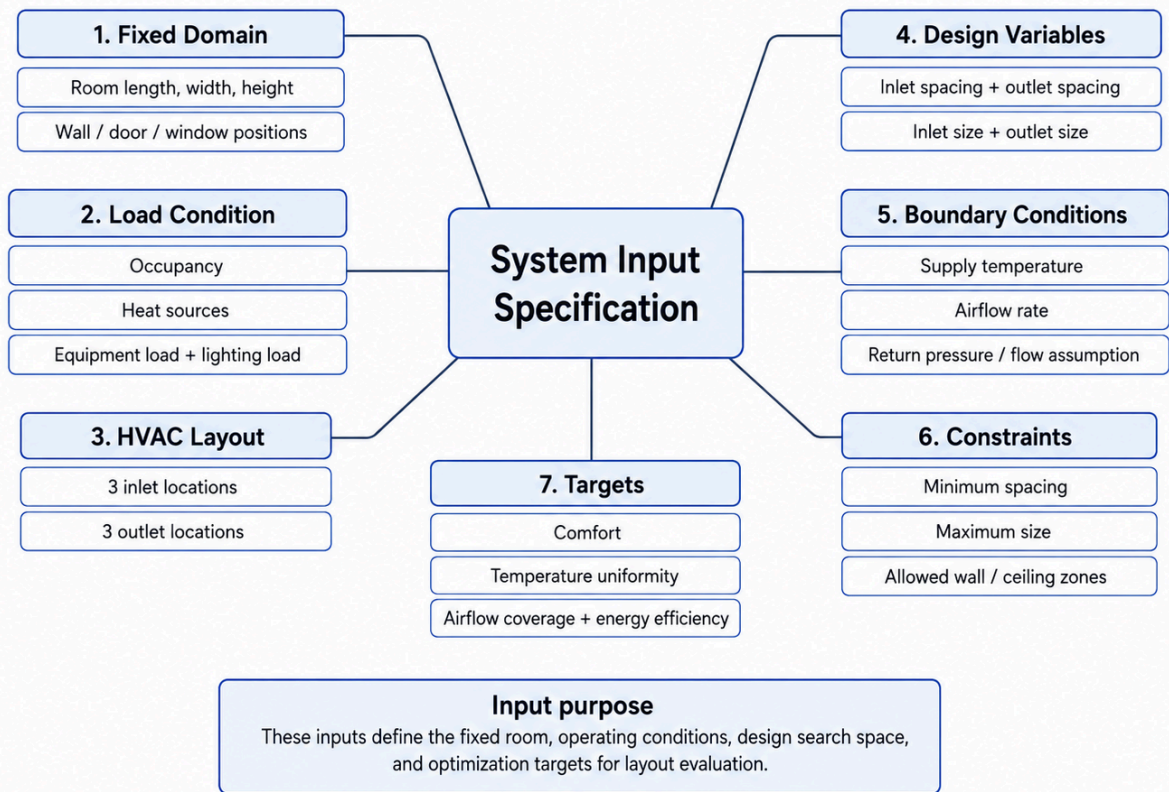
4. Target Users

User	Main Need
HVAC engineer	Quickly identify good inlet/outlet arrangements
CFD/simulation engineer	Reduce unnecessary detailed simulations
Building designer	Compare practical design options
Facility owner	Improve comfort and reduce energy use

5. Input Specification

2. Input Specification Diagram

HVAC-X Fast Layout Optimizer



Input Group

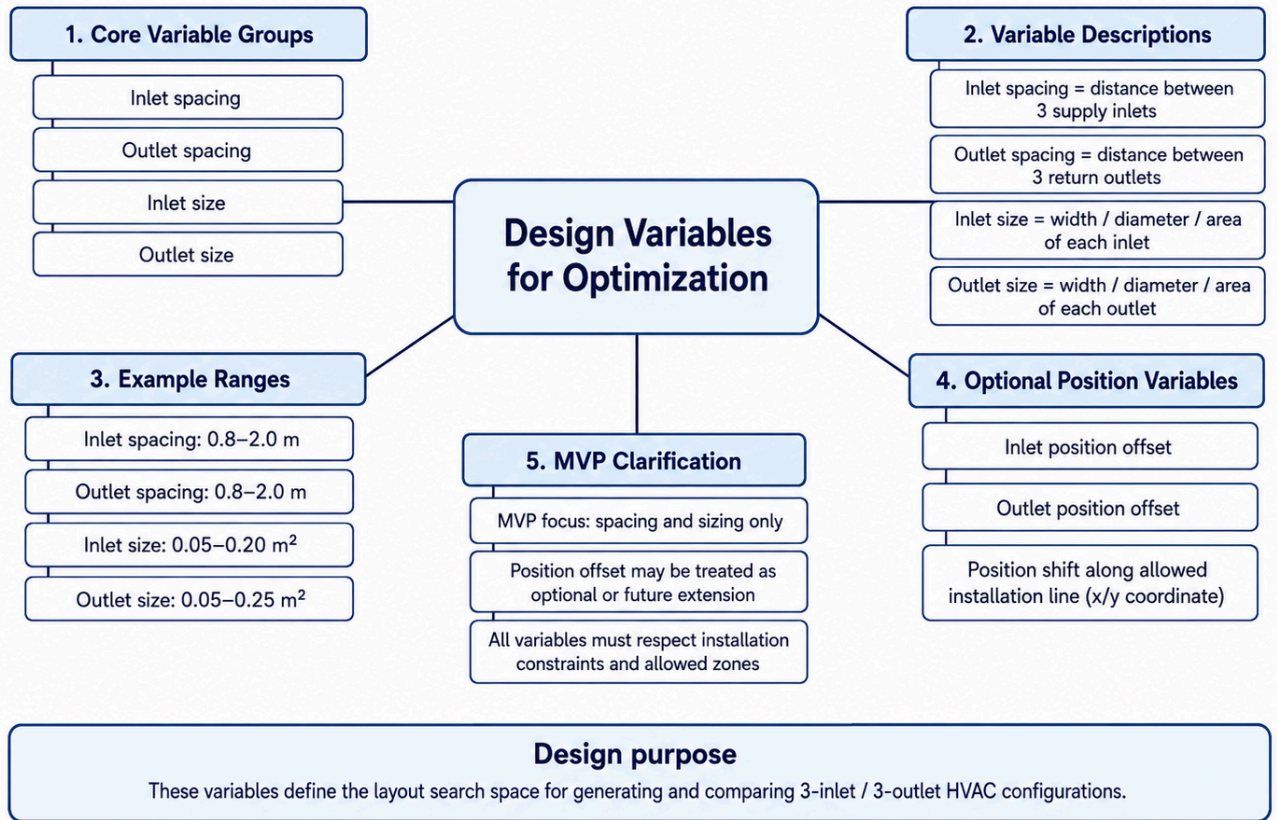
Required Input

Fixed domain	Room length, width, height, wall/door/window position
Load condition	Occupancy, heat sources, equipment load, lighting load
HVAC layout	3 inlet locations, 3 outlet locations
Design variables	Inlet spacing, outlet spacing, inlet size, outlet size
Boundary conditions	Supply temperature, airflow rate, return pressure/flow assumption
Constraints	Minimum spacing, maximum size, allowed wall/ceiling zones
Targets	Comfort, temperature uniformity, airflow coverage, energy efficiency

6. Design Variables

3. Design Variable Diagram

HVAC-X Fast Layout Optimizer



Variable	Description	Example
Inlet spacing	Distance between 3 supply inlets	0.8–2.0 m
Outlet spacing	Distance between 3 return outlets	0.8–2.0 m
Inlet size	Width/diameter/area of each inlet	0.05–0.20 m ²
Outlet size	Width/diameter/area of each outlet	0.05–0.25 m ²
Inlet position offset	Position shift along allowed installation line	x/y coordinate
Outlet position offset	Position shift along allowed installation line	x/y coordinate

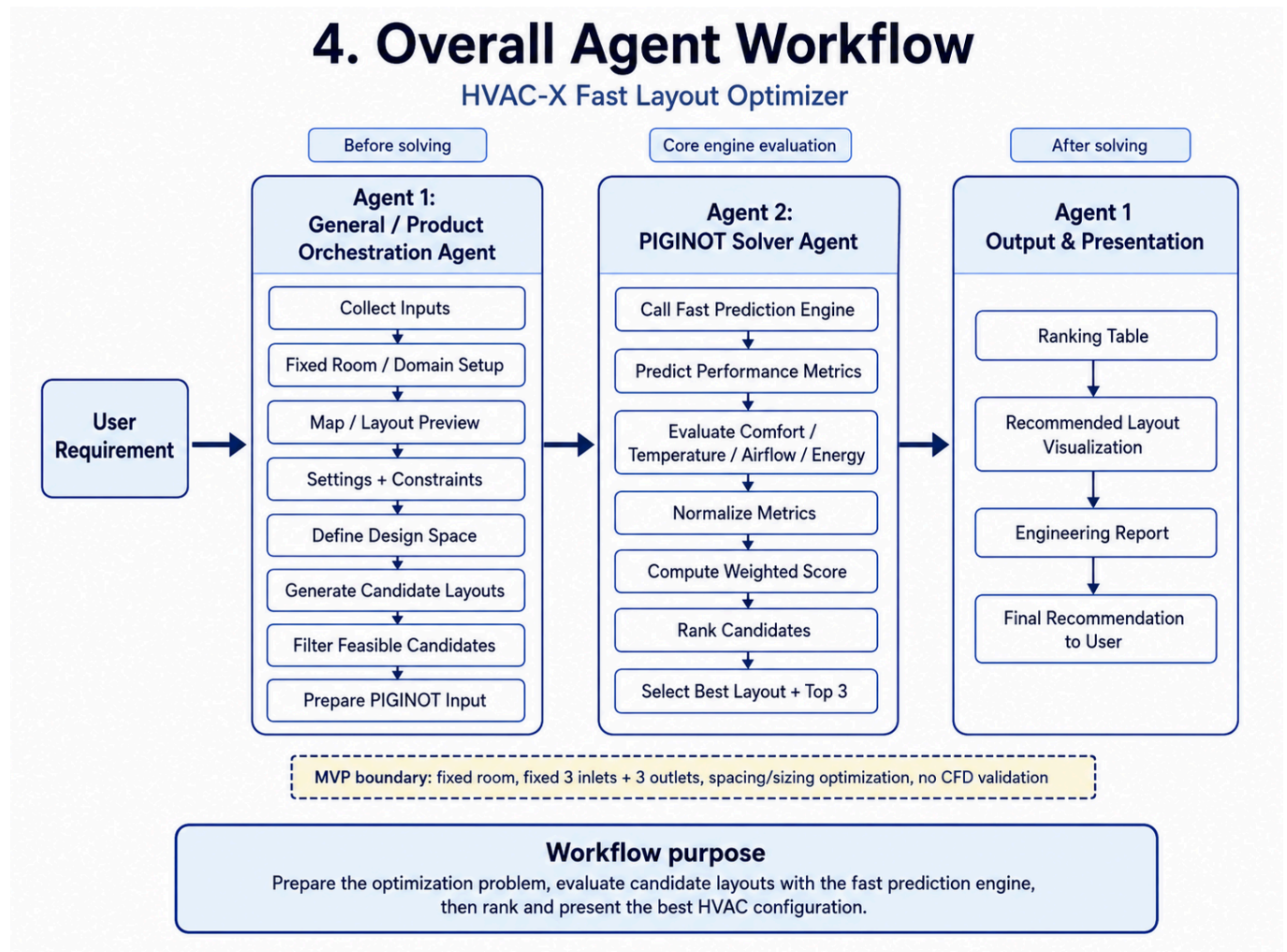
7. Output Specification

Output

Description

Structured design brief	Fixed room, load, constraint, and target definition
Candidate layout set	Generated combinations of 3 inlets and 3 outlets
Prediction results	Estimated airflow, temperature, comfort, and energy indicators
Ranking table	Candidate score based on multi-objective criteria
Best configuration	Recommended inlet/outlet spacing and sizing
Design report	Summary of assumptions, best layout, and performance comparison

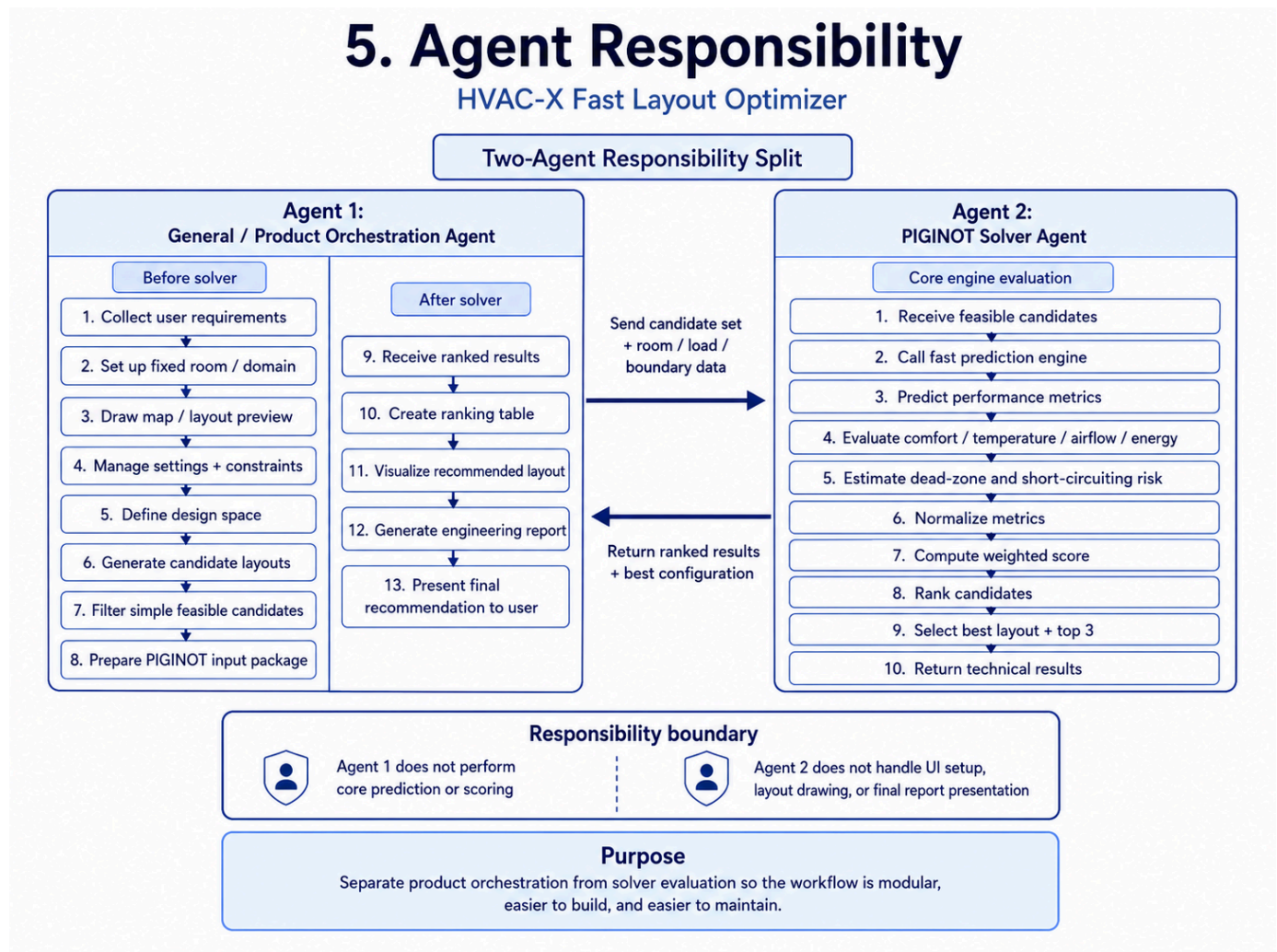
8. Agent Workflow



User Requirement
→ Fixed Domain Setup Agent

- Design Space Agent
- Candidate Generation Agent
- Fast Prediction Agent
- Ranking Agent
- Report Agent

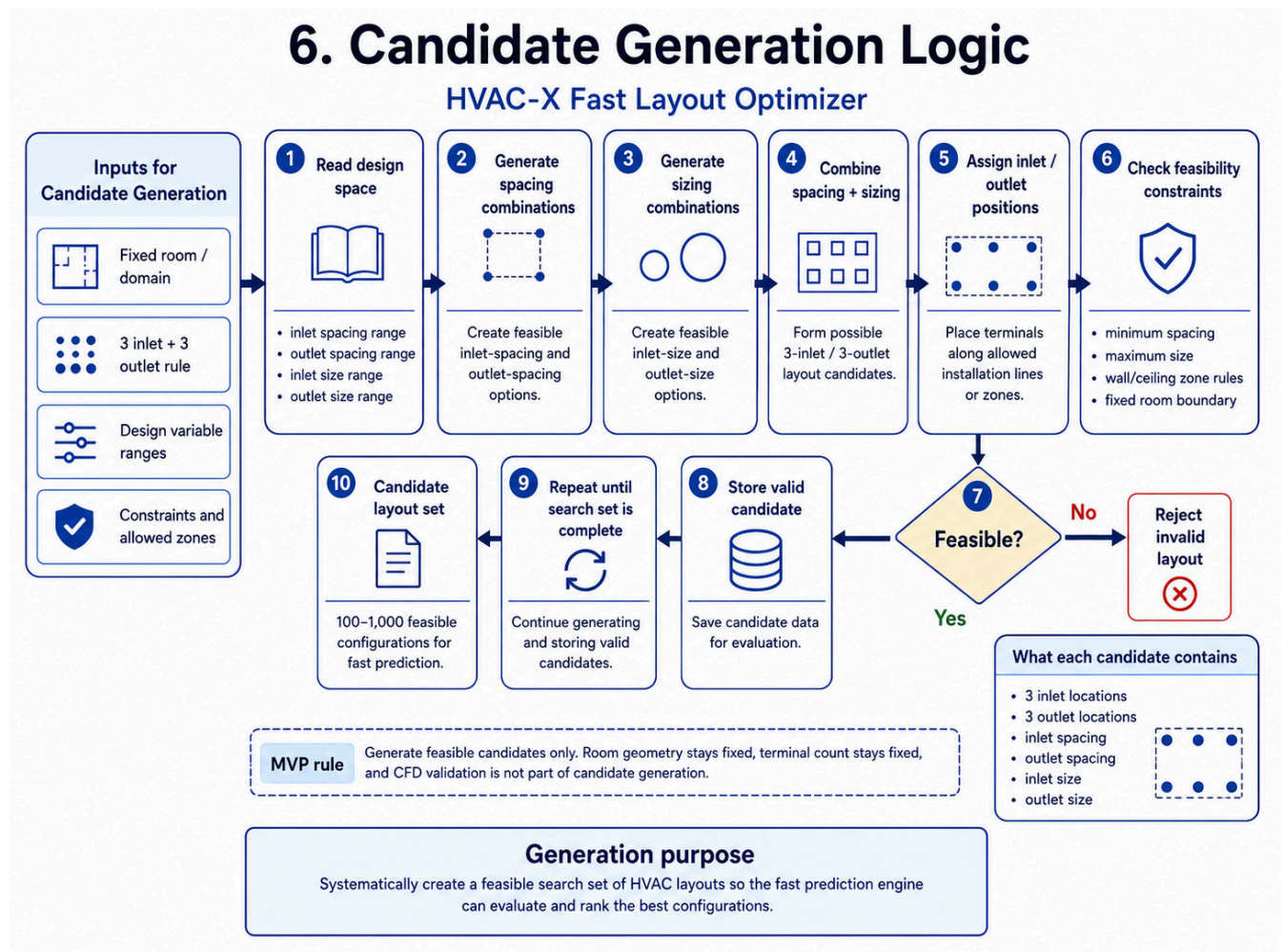
9. Agent Specification



Agent	Role	Input	Output
Fixed Domain Setup Agent	Defines fixed room and boundary conditions	Room geometry, load, HVAC assumptions	Structured design brief
Design Space Agent	Defines spacing and sizing ranges	Design brief, constraints	Design variable table

Candidate Generation Agent	Creates multiple 3-inlet/3-outlet layouts	Variable ranges	Candidate configurations
Fast Prediction Agent	Estimates performance without CFD validation	Candidate layouts, surrogate model	Predicted metrics
Ranking Agent	Scores and ranks candidates	Prediction results	Top-ranked configurations
Report Agent	Generates final design summary	Ranking results	Recommended layout report

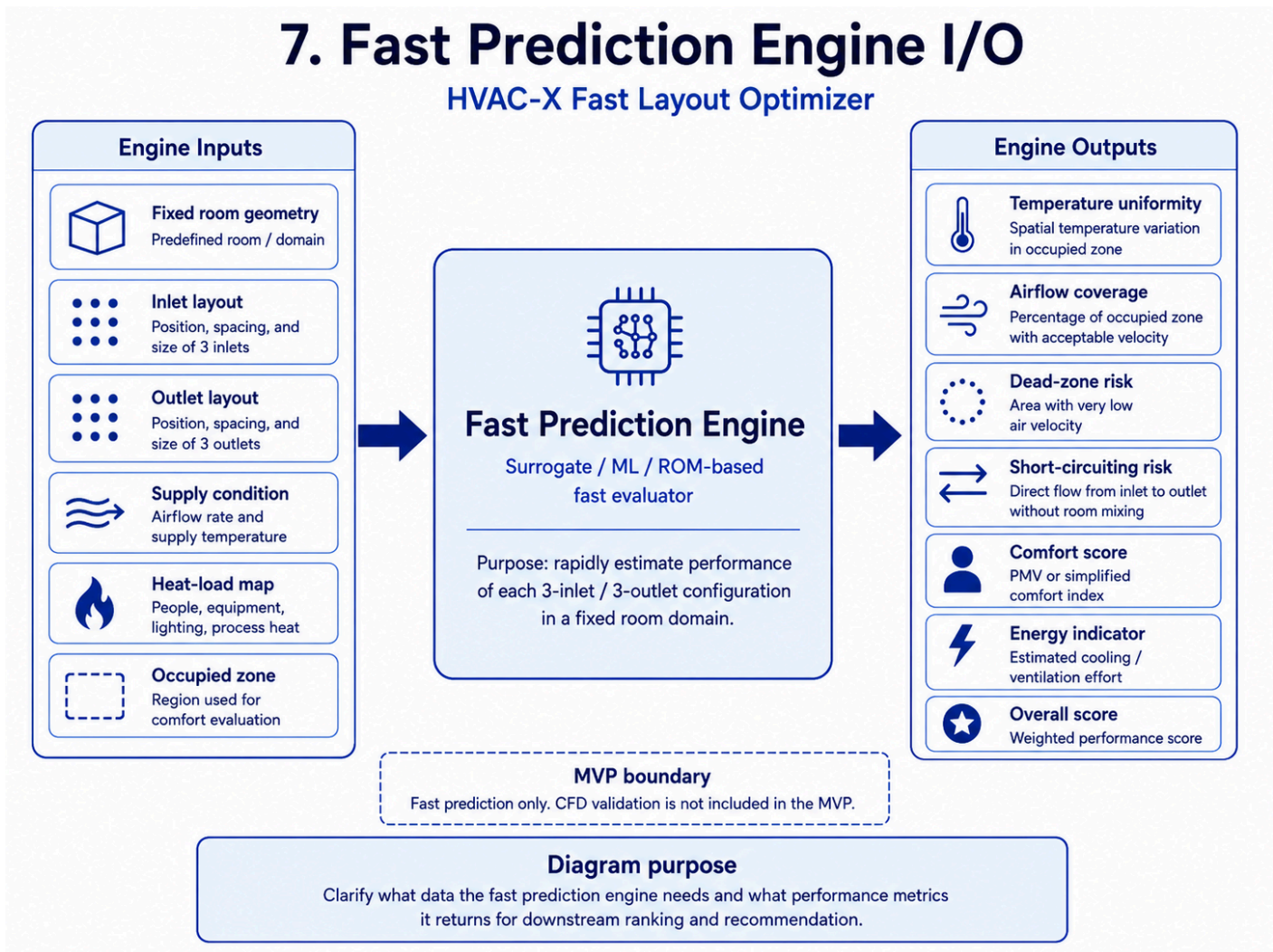
Candidate Generation Workflow:



10. Fast Prediction Engine Specification

Purpose

Rapidly estimate the performance of each 3-inlet/3-outlet configuration in a fixed room domain.



Input

Input	Description
Fixed room geometry	Predefined room/domain
Inlet layout	Position, spacing, size of 3 inlets
Outlet layout	Position, spacing, size of 3 outlets
Supply condition	Airflow rate and supply temperature

Heat-load map	People, equipment, lighting, process heat
Occupied zone	Region used for comfort evaluation

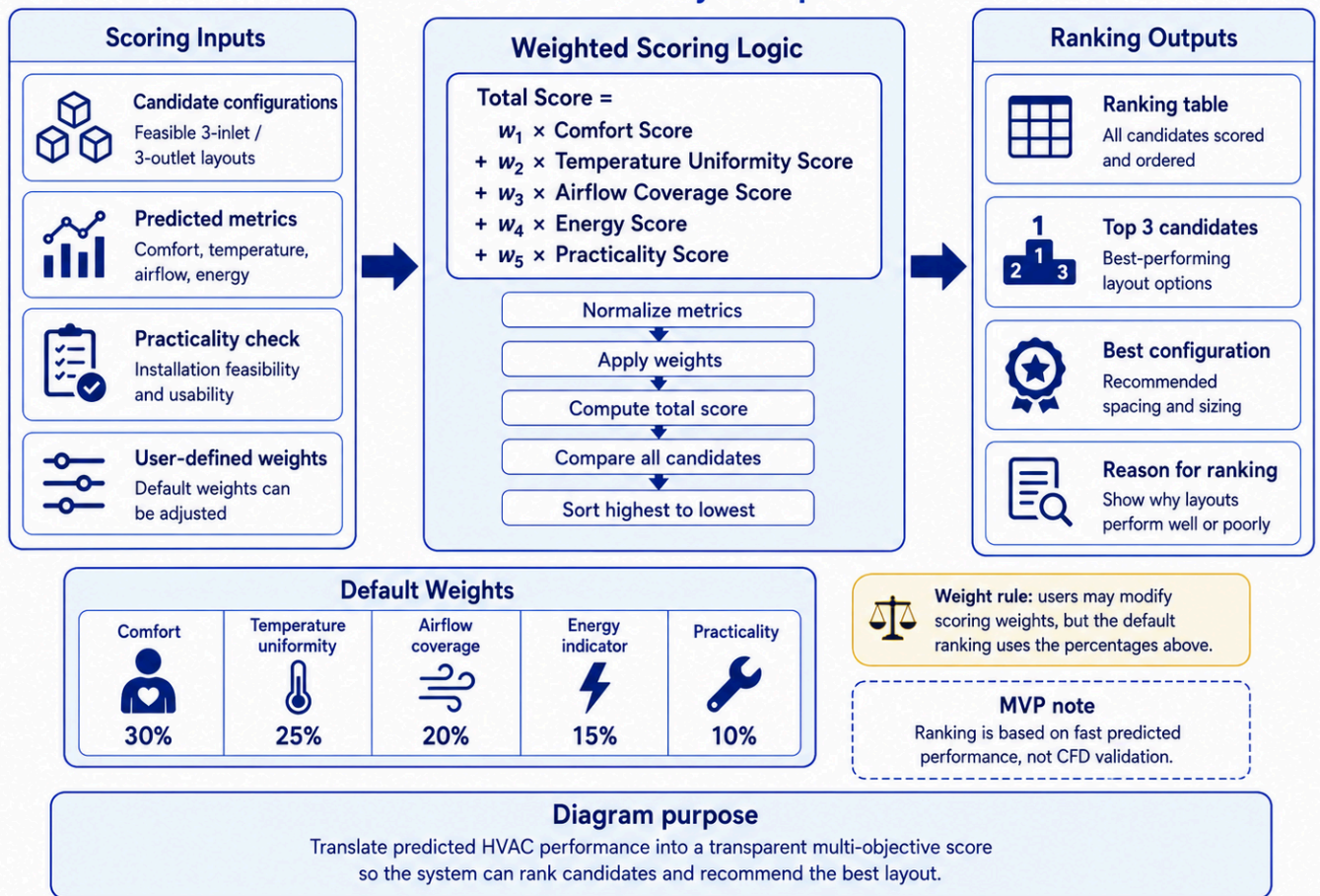
Output

Output Metric	Description
Temperature uniformity	Spatial temperature variation in occupied zone
Airflow coverage	Percentage of occupied zone with acceptable velocity
Dead-zone risk	Area with very low air velocity
Short-circuiting risk	Direct flow from inlet to outlet without room mixing
Comfort score	PMV or simplified comfort index
Energy indicator	Estimated cooling/ventilation effort
Overall score	Weighted performance score

11. Ranking Logic

8. Ranking & Scoring

HVAC-X Fast Layout Optimizer



The system ranks each candidate using a weighted score:

$$\begin{aligned} \text{Total Score} = & \\ & w_1 \times \text{Comfort Score} \\ & + w_2 \times \text{Temperature Uniformity Score} \\ & + w_3 \times \text{Airflow Coverage Score} \\ & + w_4 \times \text{Energy Score} \\ & + w_5 \times \text{Practicality Score} \end{aligned}$$

Default Weights

Metric	Weight
Comfort	30%

Temperature uniformity	25%
Airflow coverage	20%
Energy indicator	15%
Practicality	10%

12. Functional Requirements

10. Requirement Traceability

HVAC-X Fast Layout Optimizer

FR ID	Functional Requirement	Primary Module / Component	Primary Owner
FR-01	Accept fixed room / domain as input	Input Specification + Fixed Domain Setup	Agent 1
FR-02	Keep number of inlets fixed at 3	MVP Scope Rule + Candidate Generation Control	Agent 1
FR-03	Keep number of outlets fixed at 3	MVP Scope Rule + Candidate Generation Control	Agent 1
FR-04	Vary inlet/outlet spacing and sizing	Design Variable Definition	Agent 1
FR-05	Generate feasible candidate configurations	Candidate Generation Logic	Agent 1
FR-06	Evaluate candidates using fast prediction engine	Fast Prediction Engine	Agent 2
FR-07	Rank candidates using multi-objective criteria	Ranking & Scoring Logic	Agent 2
FR-08	Recommend best inlet / outlet spacing and sizing	Best Configuration Selection	Agent 2
FR-09	Generate concise engineering report	Output / Report Assembly	Agent 1

Traceability by Agent

- **Agent 1** owns setup, scope control, design space, candidate generation, and reporting
- **Agent 2** owns prediction, scoring, ranking, and best-layout recommendation

Related Diagram Links

- FR-01 ↔ Diagram 2, Diagram 4
- FR-02 / FR-03 ↔ Diagram 1, Diagram 6
- FR-04 ↔ Diagram 3, Diagram 6
- FR-05 ↔ Diagram 4, Diagram 6
- FR-06 ↔ Diagram 4, Diagram 7
- FR-07 ↔ Diagram 4, Diagram 8
- FR-08 ↔ Diagram 8, Diagram 9
- FR-09 ↔ Diagram 4, Diagram 9

MVP note

All functional requirements are traced within the MVP scope: fixed room, fixed 3 inlets + 3 outlets, spacing/sizing optimization, and no CFD validation.

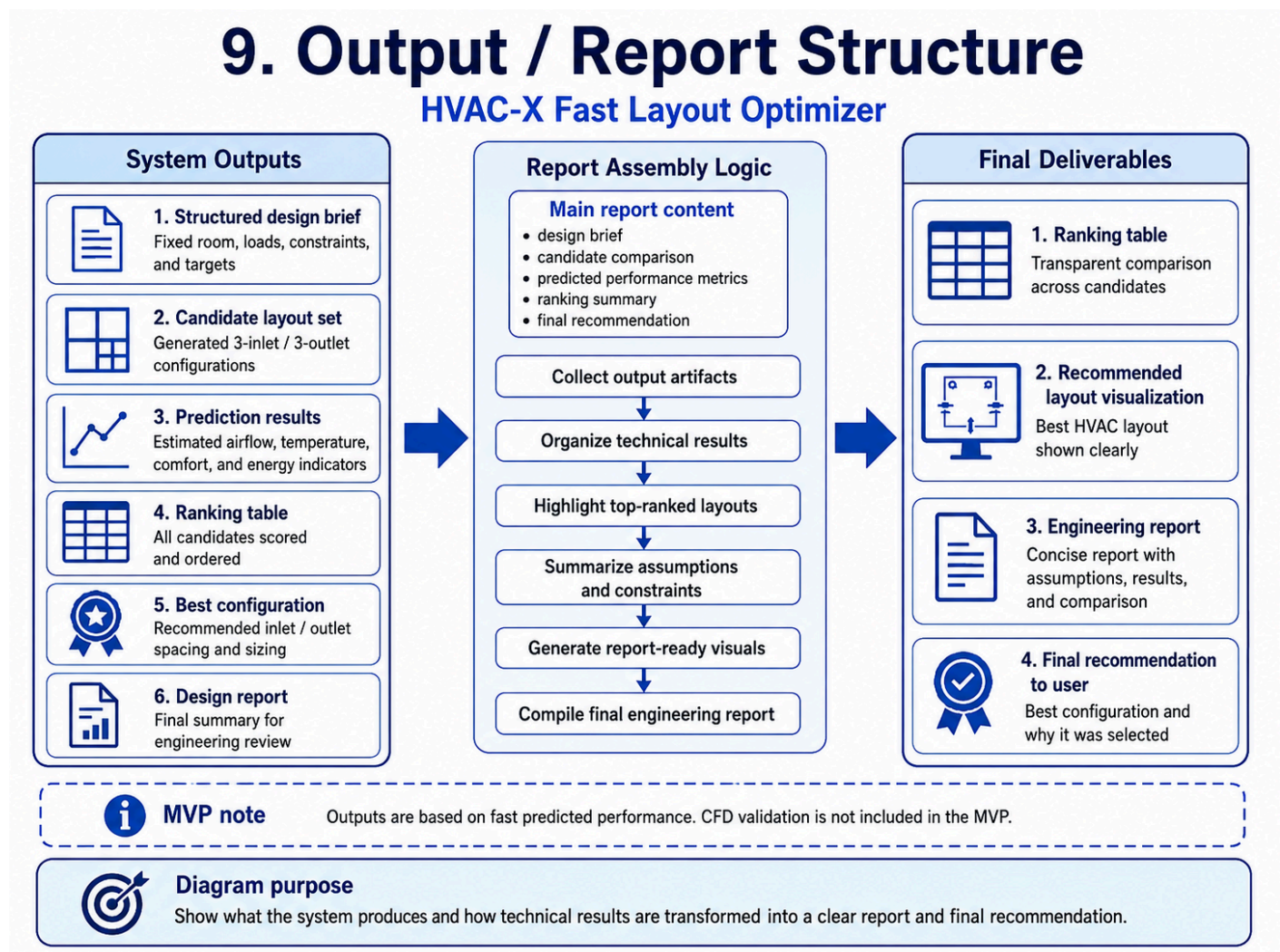
Diagram purpose

Show how each functional requirement maps to a concrete module and responsible agent so implementation can be tracked clearly.

ID	Requirement
FR-01	The system shall accept a fixed room/domain as input.
FR-02	The system shall keep the number of inlets fixed at 3.

- FR-03 The system shall keep the number of outlets fixed at 3.
- FR-04 The system shall vary inlet spacing, outlet spacing, inlet size, and outlet size.
- FR-05 The system shall generate feasible candidate configurations.
- FR-06 The system shall evaluate candidates using a fast prediction engine.
- FR-07 The system shall rank candidates using multi-objective criteria.
- FR-08 The system shall recommend the best inlet/outlet spacing and sizing.
- FR-09 The system shall generate a concise engineering report.

Output / Report Structure:



13. Non-Functional Requirements

Category	Requirement
Speed	Screen 100–1,000 candidates within minutes
Simplicity	No CFD validation required in MVP
Explainability	Ranking must show why each candidate performs well or poorly
Flexibility	User can modify spacing, sizing, and scoring weights
Reusability	Same workflow can be reused for office, lab, classroom, or industrial room
Practicality	Generated layouts must satisfy installation constraints

14. MVP Scope

Item	MVP Specification
Domain	Fixed rectangular or simplified room
Air terminals	3 inlets and 3 outlets
Variables	Spacing and sizing only
Prediction model	Surrogate/ML/ROM-based fast evaluator
Candidate number	100–1,000 configurations
Final output	Ranked table and recommended layout
Validation	Not included in MVP

15. Success Metrics

KPI	Target
Candidate screening speed	100+ configurations within minutes
Design variables handled	Inlet/outlet spacing and sizing
Best layout identification	Top 3 candidates clearly ranked

Comfort improvement	Better temperature and airflow coverage than baseline
Energy indicator improvement	Lower estimated cooling/ventilation effort
Workflow simplicity	No manual CFD setup required

16. Final Product Statement

HVAC-X Fast Layout Optimizer is an agentic AI tool for early-stage HVAC layout optimization in a fixed room domain. It keeps the room geometry fixed and evaluates multiple configurations of 3 inlets and 3 outlets by varying only inlet/outlet spacing and sizing. A fast prediction engine estimates comfort, airflow quality, temperature uniformity, and energy indicators, enabling rapid multi-objective ranking and final layout recommendation without CFD validation in the MVP.